Preface

what this book is about,
and how to read it

For thirty years I have been participating in a great quest: a quest to understand a legacy bequeathed by Albert Einstein to future generations—his relativity theory and its predictions about the Universe—and to discover where and how relativity fails and what replaces it.

This quest has led me through labyrinths of exotic objects: black holes, white dwarfs, neutron stars, singularities, gravitational waves, wormholes, time warps, and time machines. It has taught me epistemology: What makes a theory "good"? What transcending principles control the laws of nature? Why do we physicists think we know the things we think we know, even when technology is too weak to test our predictions? The quest has shown me how the minds of scientists work, and the enormous differences between one mind and another (say, Stephen Hawking's and mine) and why it takes many different types of scientists, each working in his or her own way, to flesh out our understanding of the Universe. Our quest, with its hundreds of participants scattered over the globe, has helped me appreciate the international character of science, the different ways the scientific enterprise is organized in different societies, and the intertwining of science with political currents, especially Soviet/American rivalry.

This book is my attempt to share these insights with nonscientists, and with scientists who work in fields other than my own. It is a book of interlocking themes held together by a thread of history: the history of our struggle to decipher Einstein's legacy, to discover its seemingly outrageous predictions about black holes, singularities, gravitational waves, wormholes, and time warps.